Picture pan

The original image src is converted to the target image dst through the transformation matrix M.

dst(x, y) = src(M11x + M12y+M13, M21x+M22y+M23)

If the original image src is moved 200 pixels to the right and 100 pixels downward, the corresponding relationship is.

```
dst(x, y) = src(x+200, y+100)
```

Complete the above expression:

dst(x, y) = src(1·x + 0·y + 200, 0·x + 1·y + 100)

According to the above expression, the value of each element in the corresponding conversion matrix M can be determined as.

M11=1

M12=0

M13=200

M21=0

M22=1

M23=100

Substituting the above values into the transformation matrix M, we will get:

$$M = \begin{bmatrix} 1 & 0 & 200 \\ 0 & 1 & 100 \end{bmatrix}$$

Directly use the transformation matrix M to call the function cv2.warpAffine() to complete the image translation.

Code path:

```
~/dofbot_ws/src/dofbot_opencv/scripts/2.Transform/03_move_pic.ipynb
```

```
import cv2
import   numpy as np
img =   cv2.imread('yahboom.jpg',1)
#cv2.imshow('src',img)
imgInfo = img.shape
height =   imgInfo[0]
width =   imgInfo[1]
####
matShift =   np.float32([[1,0,200],[0,1,100]])# 2*3
dst =   cv2.warpAffine(img, matShift, (height, width))    #1 data 2 mat 3 info
#   Shift Matrix
#   cv2.imshow('dst',dst)
#   cv2.waitKey(0)
```

Display the original image and the translated image comparison in jupyterLab.

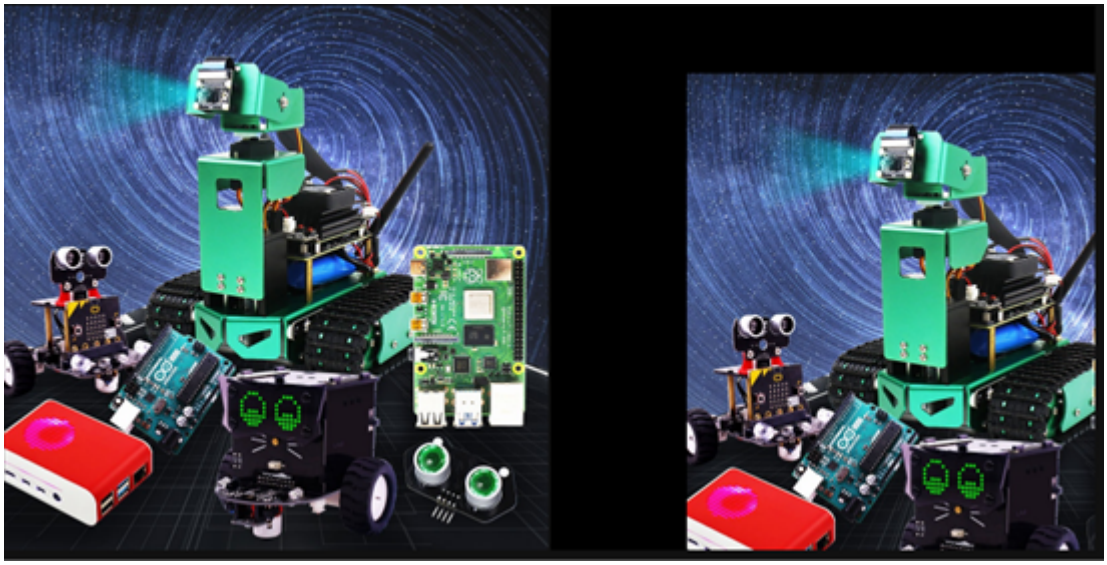
```
#bgr8 to jpeg format
import enum
import cv2
def   bgr8_to_jpeg(value, quality=75):
    return bytes(cv2.imencode('.jpg',   value)[1])
```

```
import   ipywidgets.widgets as widgets
image_widget1   = widgets.Image(format='jpg', )
image_widget2   = widgets.Image(format='jpg', )
# create a   horizontal box container to place the image widget next to
eachother
image_container   = widgets.HBox([image_widget1, image_widget2])
# display   the container in this cell's output
display(image_container)
#display(image_widget2)

img1 =   cv2.imread('yahboom.jpg',1)

image_widget1.value   = bgr8_to_jpeg(img1)

image_widget2.value   = bgr8_to_jpeg(dst)
```



As can be seen from the image, the picture has moved to the lower right corner by (200, 100).